Saumya Chaudhary

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EDUCATION

Master of Science in **Computer Science** | *University of Illinois Chicago* **August 2024 - May 2026**
Coursework: Artificial Intelligence-I, Introduction to Data Science, Big Data Mining, Responsible AI, Artificial Intelligence II, Query Processing Database Systems, Natural Language Processing, Advanced Machine Learning, Edge Computing GPA: 4.0

Bachelor of Technology in **Computer Science** | *Medicaps University* **August 2020 - May 2024**
Coursework: Software Engineering, Computer Networks, Data Structure, Object-Oriented Programming GPA: 3.7

TECHNICAL SKILLS

Programming Languages: Python, C#, JavaScript
Artificial Intelligence: Claude Code, LangChain, LangGraph, LangSmith, Autogen, Embeddings, Prompt Engineering, HuggingFace, LoRA, RAG
Libraries and Frameworks: REST API, ASP.NET Core, React, Flask, Agile Framework, Apache Spark, NGINX, Apache Kafka, TensorFlow, Keras, Azure AI, Chart.js, Microservices,
Tools & Platforms: Docker, Kubernetes, ElasticSearch**,** AWS, Git, GitLab, GitHub, Postman, Linux, CUDA, MLflow, Mosquitto, Power BI
Databases: Microsoft SQL Server, MongoDB, PostgreSQL, MySQL

EXPERIENCE

Full Stack Engineer | **School of Public Health, UIC** **April 2025 - Present**

- Developed a course management system, covering 800 + courses, with 4 stages of approval, with a complete audit trail and supporting 20-50 annual review cycles with full accountability at each stage.
- Designed a normalized relational data model in SQL Server, consolidating 25+ years of partnership and siloed contract data across 10 concentrations.
- Built a C# and JavaScript application to replace manual department-wide tracking with concentration-specific intake forms for students and faculty engagement, feeding into a unified data model generating KPIs for the school.

Software Engineering Intern | **Swastika Investmart Ltd** **January 2024 - July 2024**

- Built a scalable Flask/MongoDB backend serving real-time data for 4,500+ stocks, with Dockerized cron jobs and NGINX for scheduled tasks and API optimization.
- Resolved formatting and delivery issues across 20+ email templates and C# APIs, reducing customer setup time by 15%.
- Designed and deployed C# APIs to automate daily trade report generation for 150 stocks, accelerating research team evaluations by 30%.

Research Intern | **Space Application Centre, Indian Space Research Organization** **June 2023 - August 2023**

- Closed a critical blue band data gap across ~1M satellite images by reconstructing missing spectral data from red, green, and near-infrared channels.
- Proposed and trained a custom GAN (TensorFlow/Keras) from scratch—achieving 94% accuracy over 500 epochs and outperforming legacy blue band prediction methods.
- Integrated Dense Blocks into a U-Net backbone, surpassing the pix2pix model with a 97% R^2 score and 0.013 FID; across quantitative and visual evaluations.

PROJECTS

BankOps: Agent-to-Agent Orchestration

- Designed an agent-to-agent bank operation system with LangChain, LangGraph, Claude API, and FastAPI with an OAuth consent token for every cross-agent data request. With routing between Account, Fraud, and Loan agents, and a human-in-the-loop interrupts for card blocks and plausible fraud payments.
- Built an eval dataset comprising 63 routing correctness tests and 21 LLM as Judge rubric evaluation, achieving 83% accuracy on grounding, 60% on loan boundary, and 100% on consent and privacy, with all the runs traced through LangSmith.

StudyMate: Agentic Tutor

- Configured an agentic RAG pipeline (LangChain, LLaMA-3B) to handle multi-modal queries across PDF, DOCX, and PPT documents with context-aware intent routing.
- Formulated a FAISS semantic search pipeline with Hugging Face embeddings and recursive chunking across 500-page documents, extended with auto-generated MCQ and short-answer quizzes for comprehension evaluation.

Streaming ML Pipeline for Emergency Patient Triage

- Engineered a real-time ED triage pipeline (Kafka, XGBoost) with a 3-layer post-prediction safety system to classify patient acuity across 5 KTAS levels—processing 200 patients at 0.92 msg/s with 280ms P95 latency.
- Built a clinician feedback loop streaming override decisions through Kafka into a 5×5 model-vs-clinician matrix, persisting 50+ features per record to SQLite as auditable ground truth for retraining and LLM explainability.

CERTIFICATIONS

AWS Certified Cloud Practitioner – Foundational (Issued November 2025)